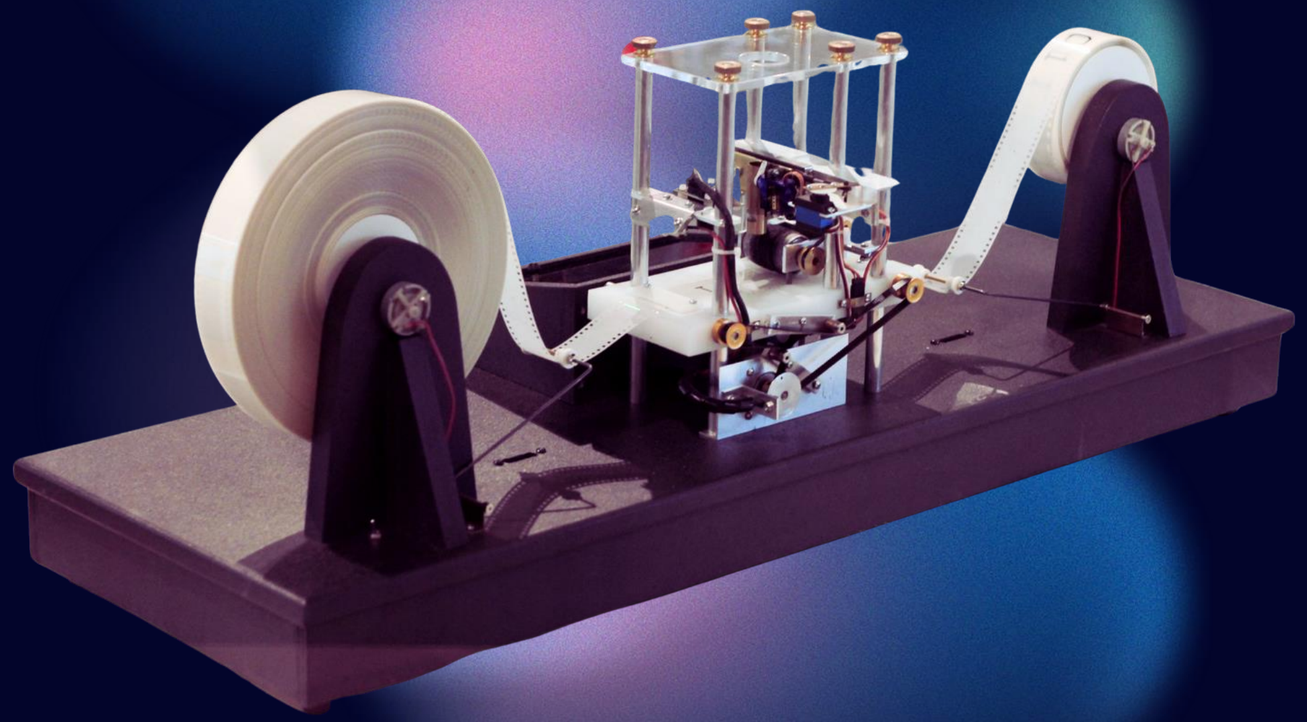
AI and Semiotics

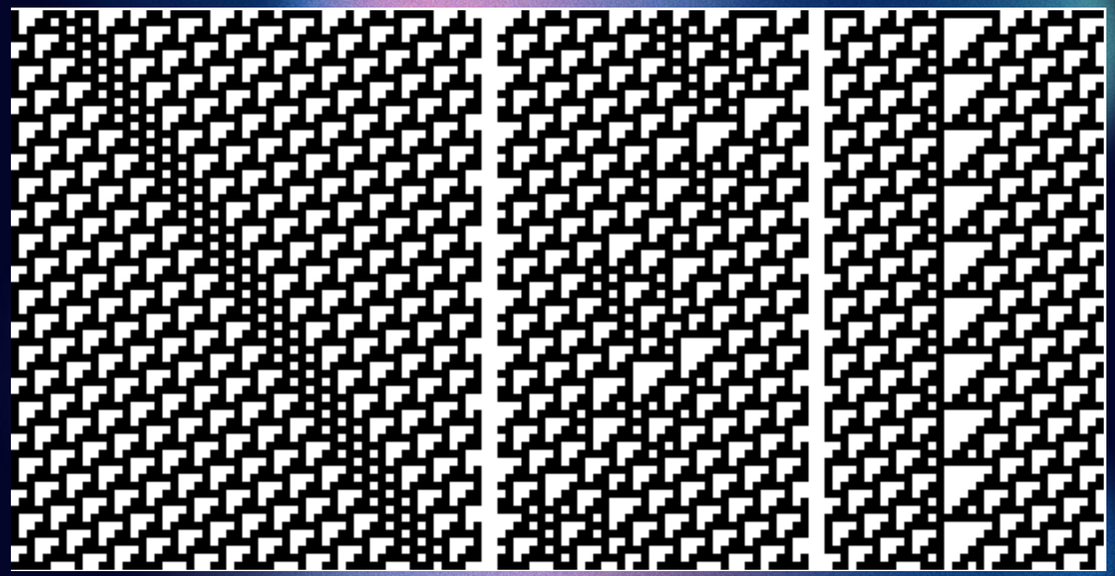
Gene Callahan
Queen Street Symposium



The Universal **Turing Machine**

In the late 1930s, Alan Turing described a remarkably simple theoretical machine—a tape, a read-write head, and a set of rules—that could read, write, and move along the tape. Despite its simplicity, this model showed how any mechanical computation could be performed.

Another **UTM**



Stephen Wolfram created the “Rule 110” cellular automaton and suspected it was Turing complete. In 2004, Matthew Cook proved that it is.

Building a Water Computer

Logically Identical

Water	Electronics
Pipes	Electrical Wires
Hot Water	High Voltage
Cold Water	Low Voltage

Building a Water Computer

Gates

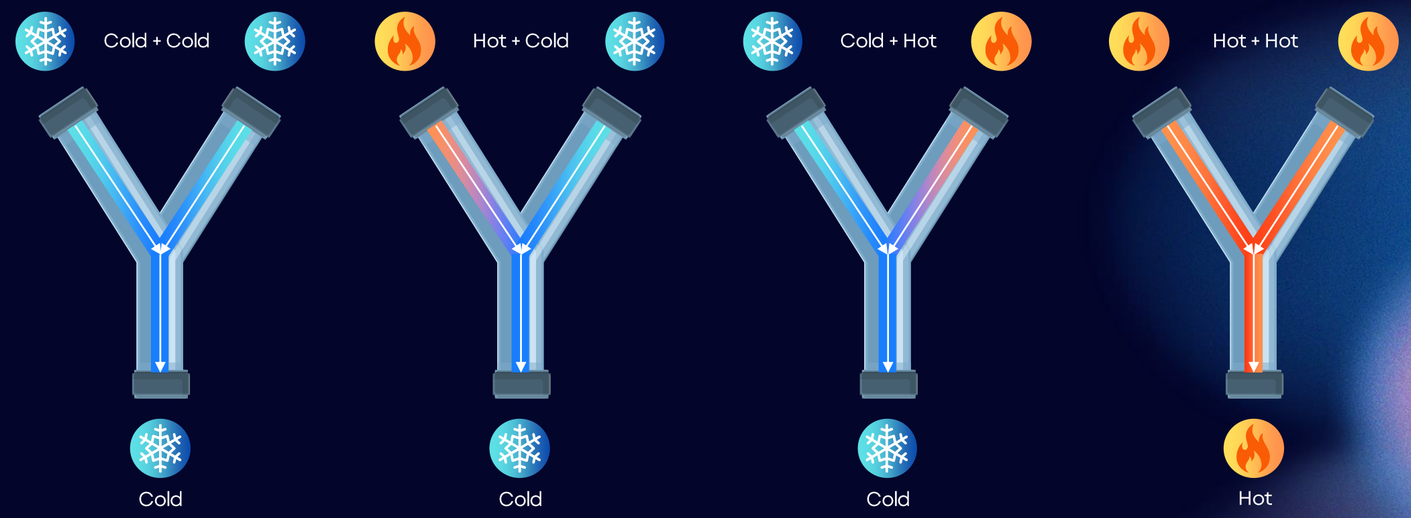
At certain points, two of our water computer's pipes meet in engineered devices called gates, which will output either hot or cold water based upon what arrives at the gate.

Building a Water Computer

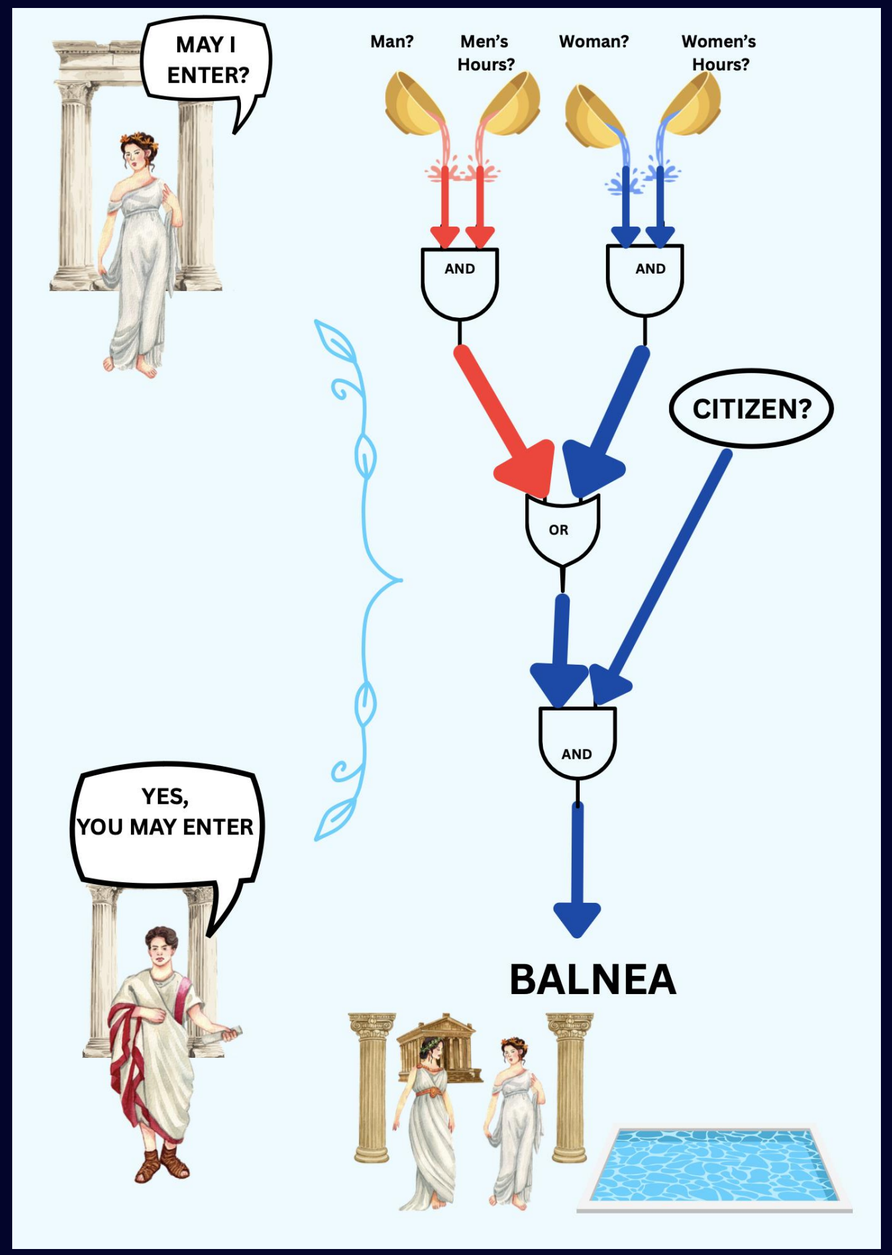
Gates

We're going to interpret hot water coming out of the gate as meaning TRUE. And we're going to say that we only want our gate to output TRUE if both pipes leading into the gate were feeding in TRUE, in other words, hot water.

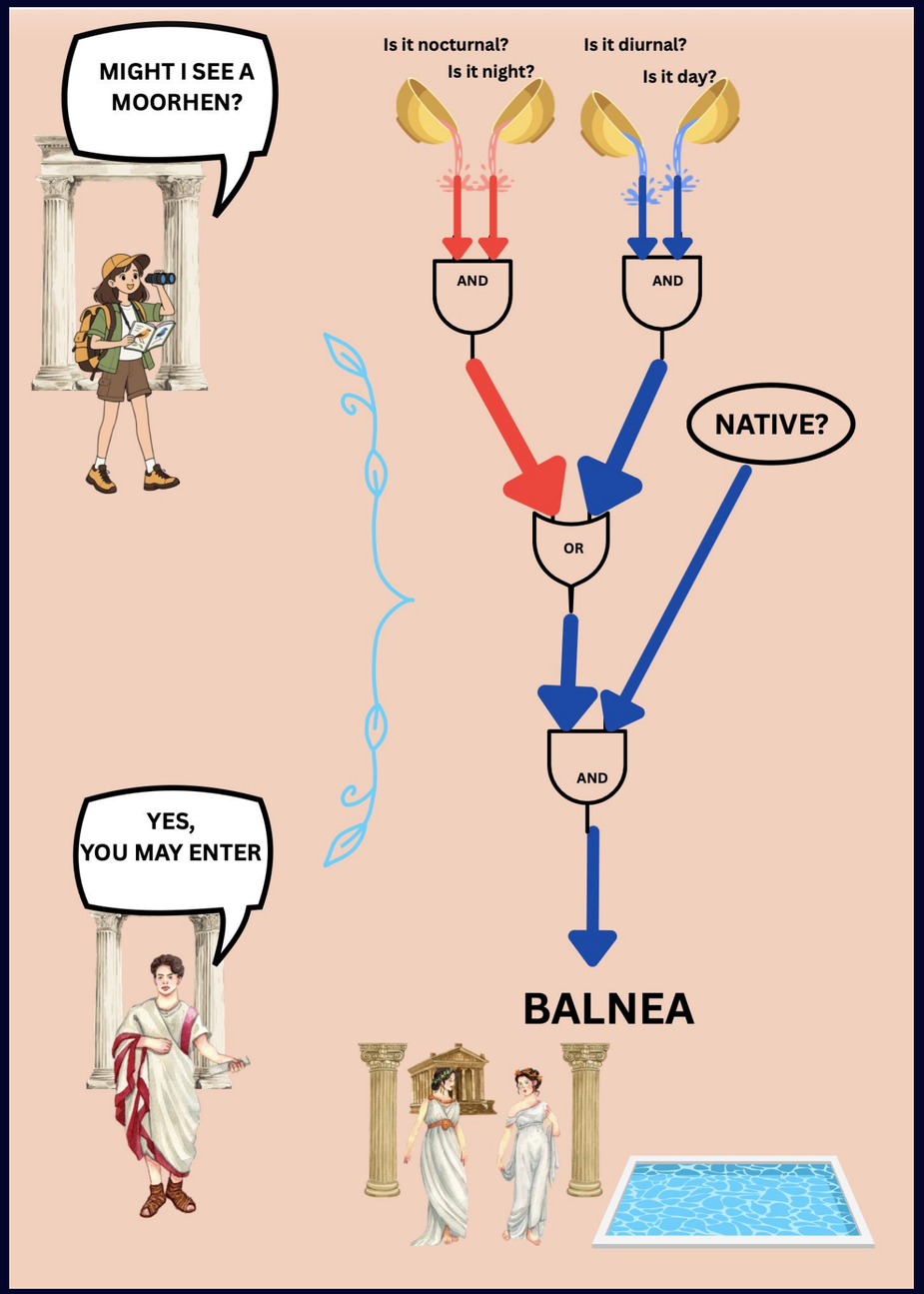
AND Gate



Let’s Answer a Question



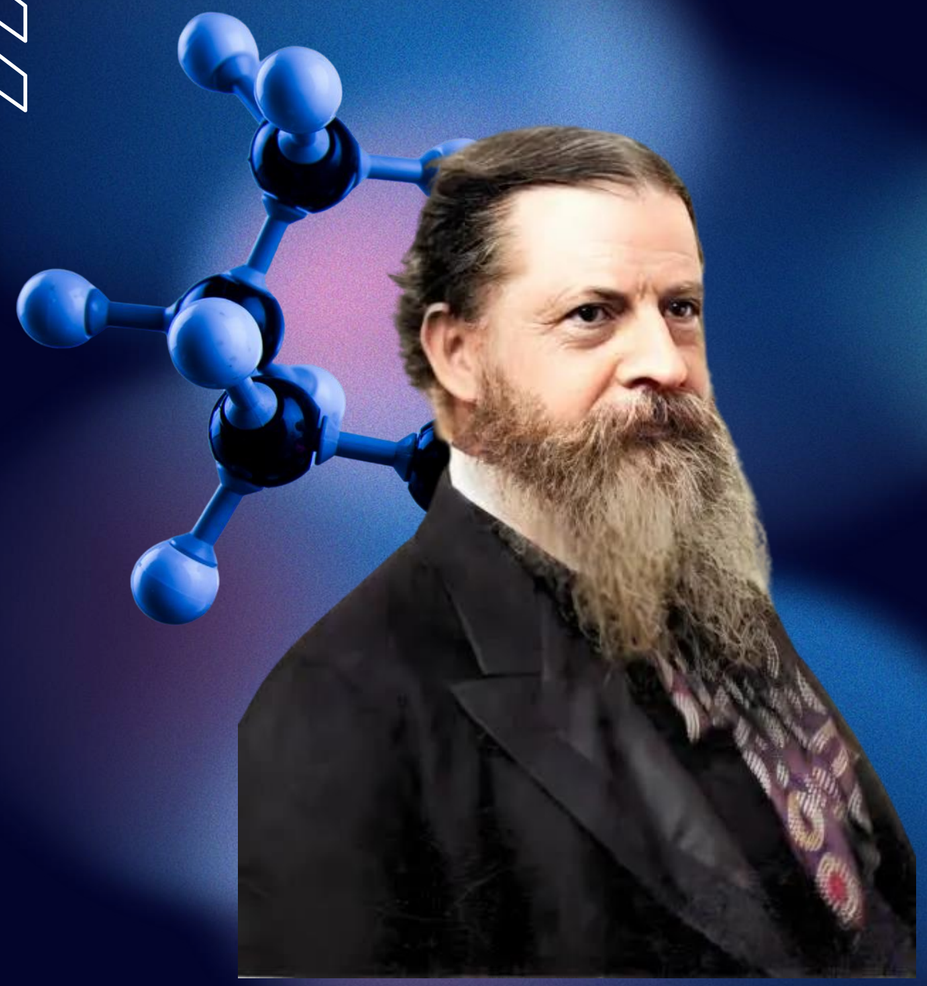
A Different Question!



All Meaning We Can Detect Is Outside the Computer

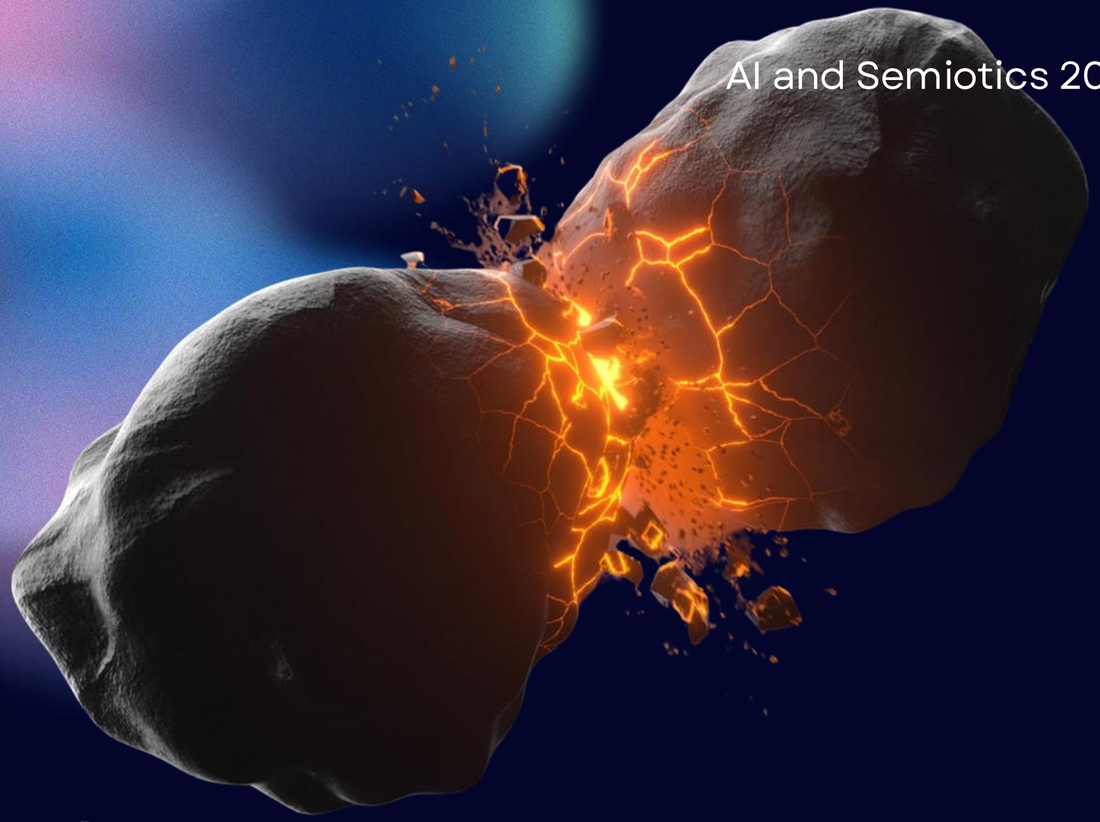
- Meaning assigned by humans
- Inputs → interpretation → outputs
- Without shared meaning: nothing

In Peircean terms, **a computing machine has no thirdness.**



C.S. Peirce

- Chemist, surveyor, mathematician, logician, astronomer and founder of modern semiotics and pragmatism.
- Bertrand Russell: "certainly the greatest American thinker ever."
- "As early as 1886, he saw that logical operations could be carried out by electrical switching circuits."



Secondness

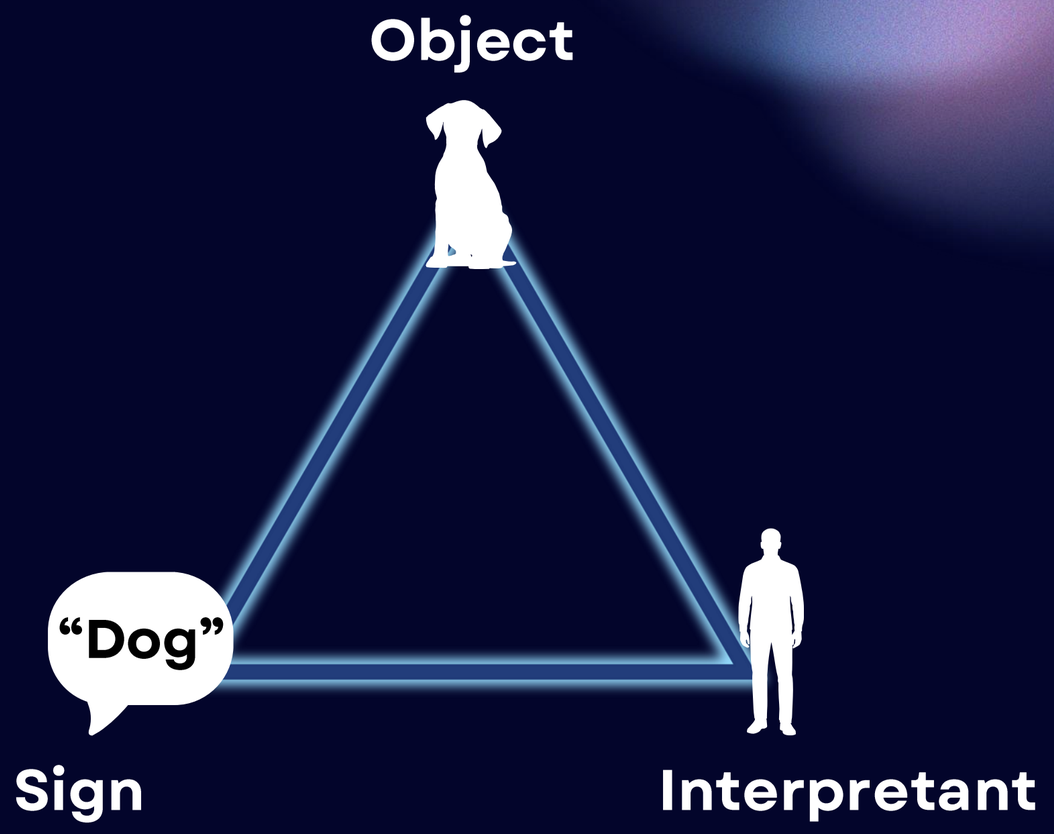
Peirce developed the concept of secondness.

When a rock tumbling down a hill meets another falling rock and deflects its course, we have a purely dyadic relationship: neither rock means anything further when hitting its companion.

Thirdness

Peirce pointed out that human language, on the other hand, is intrinsically a triadic relationship.

If I shout to you “A mind-reductionist is creeping up behind you!” I am not merely sending some sound waves to your eardrums: I am also referring to a third thing: the reductionist.



Alienation

- **Feuerbach**